

German parties shifted towards intuition-based rhetoric after the far right’s parliamentary breakthrough

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Abstract

The spread of misinformation is widely perceived as a threat to democratic deliberation, yet how political elites’ rhetorical commitments to truth shift alongside the rise of populist actors remains poorly understood. Analysing 4.5 million tweets and 59,170 parliamentary speeches by German political elites between 2015 and 2025, we measure evidence-based and intuition-based rhetoric using a validated distributed dictionary representation. Across both arenas, intuition-based language has become more prominent, and right-leaning actors consistently exhibit the lowest Evidence Minus Intuition (EMI) scores. The parliamentary entry of the extreme-right Alternative for Germany (AfD) in 2017 coincides with sharp downward shifts in EMI across the broader chamber, while a more gradual decline is observed on Twitter. These findings document an association between far-right visibility and a changing approach to truth in elite discourse in a multiparty European democracy.

1 Introduction

Democratic institutions are widely reported to be experiencing a period of erosion and declining public trust, with the spread of misinformation across social media, hyper-partisan outlets, and political polarization posing substantial challenges to democratic deliberation [1–4]. A shared consensus about what constitutes truth (e.g., public trust in elections and scientific institutions) is a necessary condition for societal coordination and, therefore, democratic functioning. Billions of people rely on social media platforms for their news consumption, which has also exposed them to a new corpus of disinformation and misinformation [3–5]. Misinformation can take many forms, ranging from conspiracy theories and propaganda to false news and deep fakes [6], and can undermine trust in scientific and academic institutions. Previous literature identifies many examples, such as the undermining of scientific findings relating to global warming and vaccinations [7, 8], or stoking of political polarization and cynicism [9, 10]. Misinformation exposure has been linked to shifts in public attitudes and behaviour, including increased electoral support for populist parties in Italy and contributions to ethnic hate crimes in Germany [11–13] leading to a breakdown in the shared consensus of truth.

In addition to the factual accuracy of specific claims, there are two critical dimensions of political discourse that are directly tied to democratic deliberation [14]. The first is *honesty*, a personal attribute whereby the speaker sincerely and authentically expresses what they deem to be true [15]. Because sincerity does not entail accuracy, a demonstrably false statement may still be perceived as honest if its speaker is seen to voice a genuinely held belief [14]. For example, Lasser et al. [16] distinguish fact- from belief-speaking in tweets by members of the US Congress, finding that belief-speaking predicts lower quality of shared news sources among Republican but not Democratic members (extending earlier evidence on partisan asymmetry in elite source sharing [17]). Belief-speaking predicts reliance on low-quality information sources, and it propagates: exposure to belief-speaking increases belief-speaking in downstream replies [18].

The second is *truth*, an external attribute concerned with how the state of the world is characterized. The *truth* may be pursued through evidence-based reasoning grounded in facts, data and other elements of external reality, or through intuition-based reasoning grounded in feelings, instincts and personal conviction [14, 19, 20]. This latter *epistemic orientation* [21] echoes the radical constructivist conception of truth that also characterised 1930s fascism, which sometimes rejected the role of evidence altogether; Nazi ideology posited an “organic truth” rooted in personal experience and accessible only through inner reflection [22, 23]. Such postures are consequential, as beliefs about how truth is to be reached shape susceptibility to misperception and misinformation [6, 19, 24].

One way to understand the role of truth in political discourse is by measuring the use of evidence verses intuition-based language in congressional speeches, by using an Evidence-Minus-Intuition (EMI) score which measures the relative prevalence of evidence-grounded rhetoric to intuition-grounded rhetoric in political text [25]. Aroyehun et al. [25] track evidence versus intuition-based language in congressional speeches,

documenting a decline in evidence-based language since the mid-1970s alongside rising polarisation, increased income inequality, and falling legislative productivity. They then extend EMI to 15 million parliamentary speech segments across seven countries and 80 years [21], finding it positively associated with deliberative democracy, transparency and net of judicial independence. When discourse relies exclusively on intuition, evidence can no longer adjudicate between competing positions, and the democratic functions of accountability, informed policy evaluation, and cross-partisan deliberation lose their footing [25]. It is important to note that evidence enables adjudication while intuition contributes moral and experiential dimensions that are often indispensable in public life. Rather, it is over-reliance on intuition at the expense of evidence that is of concern for modern democracies [21]. Most of the available evidence concerning the decline of an evidence based approach to truth in the political context originates from the United States. Whether comparable shifts occur in multiparty European democracies across arenas (Twitter and Parliament), and how they relate to the rise of new far-right actors, remains an open question.

Far right success can reshape the behaviour of mainstream parties [26–29]. Radical right electoral success exerts “contagion” effects on other parties’ immigration positions [26], and an effect that has been causally identified by exploiting electoral-threshold discontinuities in parliamentary representation [27]. During the 2015 refugee crisis, radical right parties drove mainstream parties’ attention to immigration within very short time intervals [29], and in Germany specifically, far-right actors and their issues became discursively mainstreamed in mass media debates, with a marked shift in 2015, linked to the AfD’s growing institutional access [30]. This literature has, however, focused on *what* parties talk about (issue salience) and *where* they stand (policy positions). Whether far-right success is also associated with shifts in the epistemic style of elite discourse has not been examined.

Since 2015, German politics has been reshaped by rising affective polarisation [31, 32], manifesting as positive evaluations of political ingroups and negative evaluations of opposing party supporters, especially surrounding the Alternative für Deutschland (AfD, Alternative for Germany). The AfD party, which pivoted from an anti European Union platform to anti-immigration rhetoric during the 2015 refugee crisis, entered Germany’s federal parliament, the Bundestag in 2017 [33, 34]. In 2025, Germany’s domestic intelligence agency (Bundesamt für Verfassungsschutz) classified the party as a “confirmed right-wing extremist” organization, although this designation remains subject to ongoing legal proceedings [35]. In parallel, attacks against members of established parties have risen sharply, an escalation that has been linked to an increasingly polarised political climate online and offline [12, 36–38].

Germany is an informative test case for examining how far-right parties entering parliament affect the rhetorical commitments of political elites. First, Germany’s constitutional setup differs considerably from that of the United States: a multiparty system (in which voters cast one vote for a constituency candidate and a second, decisive vote for a party list) and a post-war anti-extremist consensus that, until 2017, kept the federal Bundestag free of any party to the right of the centre-right CDU/CSU. Second, the AfD’s 2017 parliamentary entry provides one of the cleanest temporal

anchors available in any contemporary democracy: a single, well-dated event introducing a far-right actor into a chamber that had been extreme-right-free since the Nazi era. Parliamentary representation, rather than vote share alone, has been shown to constitute the theoretically decisive watershed in how established parties respond to radical right challengers, as it confers resources, media attention, and credibility as a durable competitor [27]. Third, as the European Union’s largest democracy, evidence from Germany speaks directly to the resilience of European democratic discourse more broadly: European radical right parties have repeatedly adopted the discursive repertoires of successful counterparts abroad rather than developing them independently, so an erosion of evidence-based justification in German elite discourse is unlikely to stay nationally contained [39, 40].

We ask whether the rhetorical balance between evidence- and intuition-based language in German political elite communication has shifted between 2015 and 2025, and whether such shifts coincide with the rising visibility of far-right actors. We analysed 4.5 million tweets from over 2,000 German politicians and 59,170 plenary speeches from the 18th–21st Bundestag, applying a validated Distributed Dictionary Representation (DDR) of evidence- and intuition-based language and testing for systematic differences across ideological camps and for structural shifts coinciding with the AfD’s parliamentary entry.

Our design exploits two complementary observation arenas with opposite failure modes. One arena is politicians’ rhetoric on Twitter (renamed X in 2023; we retain the original name, as it applied throughout most of our observation period), and the other is the parliament (Bundestag) itself. Twitter provides open and high-velocity communication but is subject to self-selection and algorithmic amplification. In contrast, the Bundestag is procedurally rigid and unaffected by platform-side change, but is constrained by formal speech rules. A shift towards an intuition based approach to truth visible in both arenas is therefore unlikely to be an artifact of the specific attributes of either. Ours is among the first studies to use platform data obtained through an official request under the European Union’s Digital Services Act (DSA) Article 40 (see Methods). A regulatory regime that, since 2024, has made systematic longitudinal research on very large online platforms feasible at the scale required for the present question. The result is the first cross-arena, multiparty European evidence on the relationship between far-right visibility and the rhetorical commitments to truth of political elites.

2 Results

2.1 Twitter, Bundestag speeches and ideological grouping

Twitter is an important platform for political agenda setting, with elites using it strategically through hashtag campaigns, real-time commentary on high-attention events, amplification of external movements [41] (e.g., Fridays for Future), and metacommunication about campaign developments [42, 43]. Its public-facing visibility and retweet features create spillover effects to traditional media [44], making it particularly informative for studying elite communication. We analyse a Twitter corpus of 4.5 million tweets from over 2,000 German politicians across eight parties between 2015 and 2025. We update the dataset originally collected by Lasser and colleagues [17] by retrieving

3.5 million additional tweets via the official Twitter handles of German parliamentarians elected in 2022 (GESIS handle list) [45], supplemented with the handles of the Lasser dataset. This results in a corpus of all tweets posted by every politician who was active during one or more of the given legislative periods of the German Bundestag.

Parliamentary speeches represent the most institutionalised form of political communication in Germany, governed by formal procedures and normative constraints that distinguish them from elite communication on social media. To train an embedding model fine tuned on parliamentary language, we combine two historical German Parliamentary Corpora (1867–2021) [46–48], comprising 329,390 speeches by 3,786 politicians across 22 parties. This combined corpus is used solely to pre-train the embedding model on a large body of institutional political text. Our empirical analyses draw on a separate set of publicly available Bundestag speeches collected from the official document server (see Methods), comprising 59,170 speeches from 7 parties and 1,391 politicians delivered during the 18th–21st (2015–2025) electoral terms.

Political leaning was operationalized based on the actor’s party affiliation, following the left–right positioning of the Chapel Hill Expert Survey (CHES) [49]. Accounts affiliated with the Alternative for Germany (AfD) were classified as far right, those associated with Die Linke and BSW as left-leaning, and all remaining actors as centrist. This aggregation supports analyses of the role of far-right actors relative to the rest of the political spectrum. We additionally report fine-grained party-level analyses in the results.

2.2 Temporal dynamics of evidence- and intuition-based rhetoric

To identify evidence- and intuition-based components in the texts, we apply a Distributed Dictionary Representation (DDR) approach following the pipeline of Aroyehun and colleagues [25], using two existing English intuition and evidence dictionaries developed and validated within this framework. The German versions of the dictionaries were translated, reviewed, and validated in a pre-registered online survey ($N = 47$; see Methods). The resulting keywords are shown in Extended Data Table A1. Following the DDR approach, those keywords were converted into embeddings, and the average across embeddings in each dictionary provided a single vector representation that was used in the analysis.

A gensim word2vec model was pre-trained on the historical Bundestag corpus (1867–2025) and fine-tuned separately on tweets and on contemporary Bundestag speeches to obtain platform-specific embeddings. For each text, we computed the cosine similarity between its averaged word-embedding representation and the averaged representation of each dictionary. The two resulting similarity scores per text were independently z -standardised across the corpus, and we then computed the *Evidence Minus Intuition* (EMI) score as the difference between the two: texts with $EMI < 0$ lean towards intuition-based language and texts with $EMI > 0$ towards evidence-based language. We validated EMI at the document level through a Prolific survey ($N = 284$; see Methods) in which human raters annotated a sample of 1,300 parliamentary passages stratified on both dimensions, across time and party affiliation. Using speeches that a majority of human raters agreed were representative of intuition

or evidence, we found satisfactory agreement between the computed intuition-based and evidence-based similarity scores and human ratings with an AUC of 0.72 for EMI (see Methods and Supplementary Note 2.11). Extended Data Tables A2 and A3 show examples of passages with high intuition or high evidence loadings. Robustness checks using generic German fastText embeddings yield a correlation of $r = 0.93$ between EMI scores produced by the two embedding models, and bootstrapping the dictionary composition (1,000 iterations) confirms that the document-level scores are not driven by a small subset of influential words (see Supplementary Note 2.7).

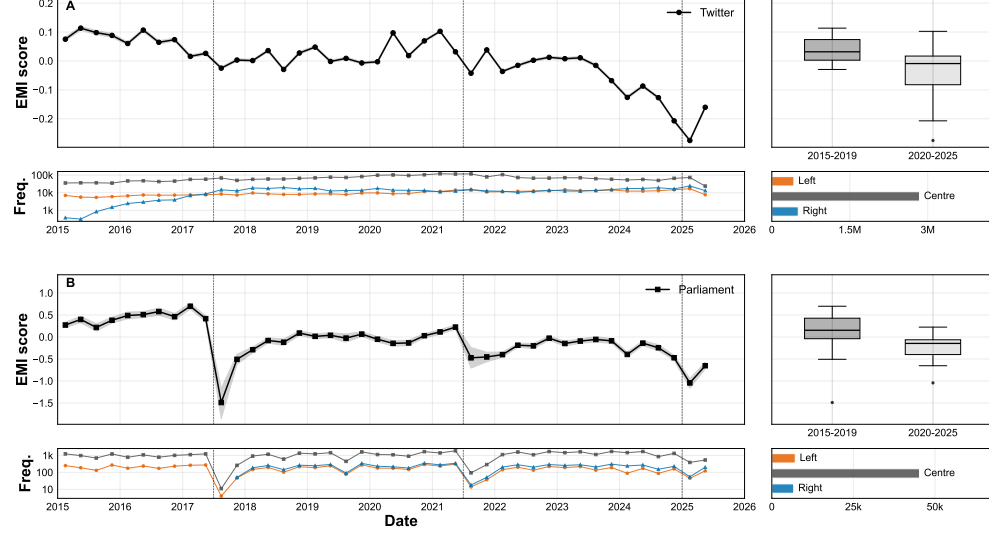


Fig. 1 Temporal trends and density distributions of evidence- and intuition-based language in tweets and parliamentary speeches. **A**, Quarterly mean EMI in tweets with log transformed frequency of tweets per leaning, **B**, Quarterly mean EMI in Parliament speeches with log transformed frequency of speeches per ideology; shaded bands are bootstrapped 95% confidence intervals (1,000 resamples). Dashed vertical lines in **A** and **B** mark the election breakpoints (1 July 2017, 1 July 2021, 1 January 2025). **C,D**, Document-level EMI scores on Twitter and in Parliament as Box plots for the periods 2015–2019 and 2020–2025. Box plots show the median (centre line), interquartile range (box bounds, 25th and 75th percentiles) and whiskers extending to the most extreme values within $1.5 \times$ the interquartile range from the box; points beyond the whiskers are plotted individually as outliers.

We computed the quarterly average similarity of intuition ($\langle D'_i \rangle$), evidence ($\langle D'_e \rangle$), and EMI ($\langle D'_{emi} \rangle$) for speech in each corpus using a 60-day rolling window. Figure 1 shows how these distributions shifted between 2015 and 2025. The 2019 cutoff falls after the AfD's initial entry into the Bundestag in late 2017 and roughly coincides with the subsequent emergence of new far-right Twitter accounts. Rhetoric scores shift towards the intuition-based end of the scale over time on both platforms. Time-series trends are shown relative to the four-yearly federal elections, which we treat as epistemic shocks. Across both arenas, EMI differs significantly between the early (2015–2019) and late (2020–2025) periods. On Twitter, mean EMI declined from 0.027

to -0.035 ($t = 78.58$, $P < 0.001$, Cohen's $d = -0.083$, 95% $CI_{\text{early}} = [0.026, 0.028]$, 95% $CI_{\text{late}} = [-0.036, -0.034]$; Mann-Whitney $U = 1.77 \times 10^{12}$, $P < 0.001$). Parliamentary speeches show a substantially larger decrease, from 0.205 to -0.147 ($t = 33.75$, $P < 0.001$, Cohen's $d = -0.281$, 95% $CI_{\text{early}} = [0.190, 0.221]$, 95% $CI_{\text{late}} = [-0.160, -0.134]$; Mann-Whitney $U = 4.92 \times 10^8$, $P < 0.001$).

2.3 Partisan patterns in evidence- and intuition-based rhetoric

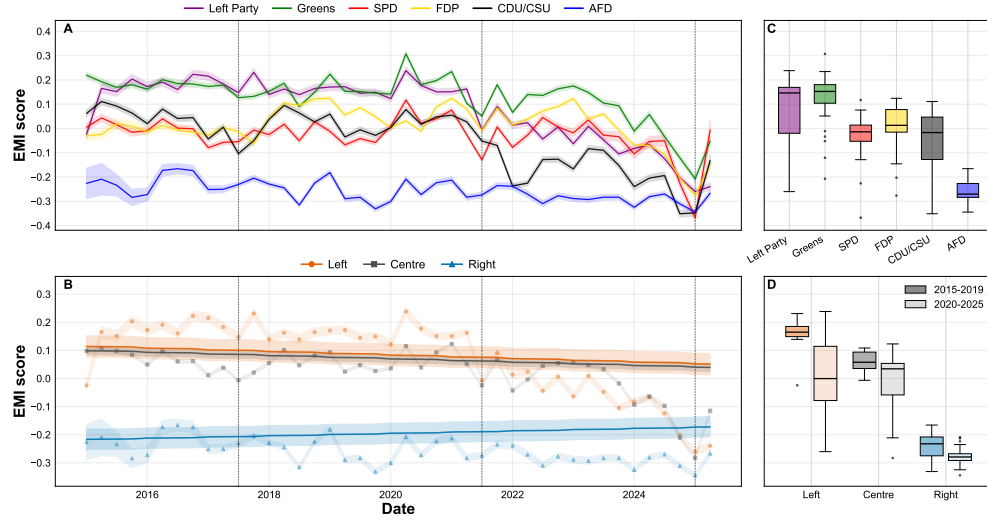


Fig. 2 Temporal dynamics of Evidence Minus Intuition (EMI) scores by political leaning on Twitter, 2015–2025. **A**, Quarterly mean EMI per party with bootstrapped 95% confidence intervals (1,000 resamples; shaded bands). **B**, Quarterly mean EMI per leaning with bootstrapped 95% confidence intervals and fitted from the linear mixed-effects model per leaning (Equation 1); shaded bands are 95% confidence intervals obtained via the delta method. Dashed vertical lines in **A** and **B** mark the model's election breakpoints (1 July 2017, 1 July 2021, 1 January 2025). **C**, Document level EMI as Boxplots per party **D**, **E**, Document level EMI as Boxplots per leaning before (**D**; 2015–2019) and from (**E**; 2020–2025) the sample split. Party abbreviations: CDU/CSU, Christian Democratic Union/Christian Social Union; SPD, Social Democratic Party of Germany; AfD, Alternative for Germany; FDP, Free Democratic Party.

Party-level mean EMI scores are reported in Extended Data Tables A4 and A5; pairwise differences are statistically significant most combinations expect Greens vs. Left Party and CDU/CSU vs. SPD (Tukey's HSD post-hoc test). A consistent ordering emerges across platforms visualized in Figures 2 and 3. On Twitter, mean EMI is highest for Alliance 90/The Greens (0.135, 95% CI [0.133, 0.136]), followed by Die Linke (0.049, 95% CI [0.047, 0.051]) and the FDP (0.011, 95% CI [0.009, 0.012]); the SPD (-0.019 , 95% CI [-0.020 , -0.017]) and CDU/CSU (-0.041 , 95% CI [-0.042 , -0.039]) show small negative values; the AfD ranks lowest by a substantial margin (-0.268 , 95% CI [-0.270 , -0.266]).

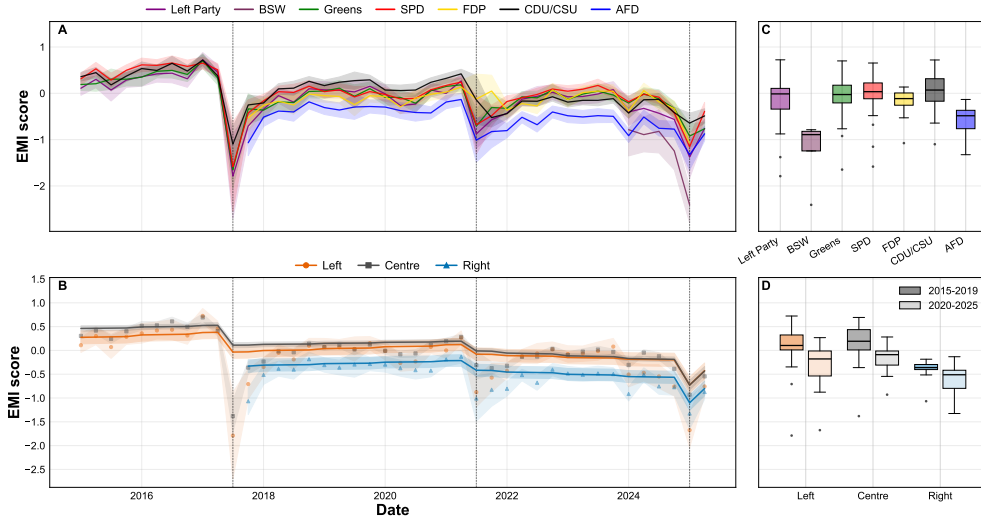


Fig. 3 Temporal dynamics of Evidence Minus Intuition (EMI) scores by political leaning in the German Bundestag, 2015–2025. **A**, Quarterly mean EMI per party with bootstrapped 95% confidence intervals (1,000 resamples; shaded bands). **B**, Quarterly mean EMI per leaning with bootstrapped 95% confidence intervals and fitted trajectories from the piecewise linear mixed-effects breakpoint model per leaning (Equation 2); shaded bands are 95% confidence intervals obtained via the delta method. Dashed vertical lines in **A** and **B** mark the model's election breakpoints (1 July 2017, 1 July 2021, 1 January 2025). **C**, Document level EMI as Boxplots per party **D**, **E**, Document level EMI as Boxplots per leaning before (**D**; 2015–2019) and from (**E**; 2020–2025) the sample split. The Bundestag's summer recess ran into the 24 September 2017 federal election. The old Bundestag held its last sitting week in early September, and the new one only constituted on 24 October, so there were almost no plenary speeches that quarter explaining the gap in the data. Party abbreviations: CDU/CSU, Christian Democratic Union/Christian Social Union; SPD, Social Democratic Party of Germany; AfD, Alternative for Germany; FDP, Free Democratic Party; BSW, Sahra Wagenknecht Alliance.

Parliamentary speeches show a similar but partly re-ordered pattern. The SPD shows the highest mean EMI (0.113, 95% CI [0.092, 0.135]), followed by CDU/CSU (0.108, 95% CI [0.089, 0.127]), Die Linke (0.051, 95% CI [0.021, 0.080]), and Alliance 90/The Greens (0.039, 95% CI [0.014, 0.065]); a negative value is observed for the FDP (−0.110, 95% CI [−0.140, −0.081]). The AfD again sits near the bottom (−0.473, 95% CI [−0.502, −0.443]); Bündnis Sahra Wagenknecht (BSW), which was founded in January 2024 and thus entered the chamber late in the observation window with a small sample of speeches ($n = 153$), shows the lowest mean overall (−1.098, 95% CI [−1.311, −0.885]). At the ideological-camp level (Extended Data Tables A4–A5), the same hierarchical ordering holds across both platforms: right-leaning actors consistently display the lowest mean EMI. Figures 2 and 3, the quarterly aggregated means (Panel A) and reveal that, over time, the centrist and left-leaning camps shift towards the lower EMI levels of the right-leaning camp. This convergence is gradual and approximately linear on Twitter, whereas in the Bundestag it is characterised

by abrupt level shifts that align temporally with changes in the composition of the Bundestag, especially with electoral entries of new parties.

To formally test these dynamics, we fit a sequence of linear mixed-effects models with random intercepts and slopes at the actor level, controlling for within-actor temporal autocorrelation via a one-quarter lagged EMI term (Methods, Equations 1–3). For parliamentary speeches we additionally introduced piecewise structural breakpoints at each German federal election (2017, 2021, 2025), first constraining the breakpoint effects to be uniform across camps and then allowing them to vary by ideology. To rule out that the estimated time and breakpoint effects reflect a shift in topic mix rather than in epistemic style, each of these specifications was additionally re-fit with document-level topic fixed effects derived from a BERTopic model (See Methods and Supplementary Note 4); the time-trend, leaning and constrained breakpoint coefficients reported below are robust to this control (See Supplementary Note 4) and EMI follows similar trends across topics with some outliers (Extended Data Figures A1 and A2).

On Twitter (Extended Data Table A6, Figure 2 Panel B), right-leaning actors exhibit substantially lower baseline EMI ($\beta = -0.111$, $P < 0.001$, 95% CI $[-0.149, -0.072]$) than the centrist reference category ($\beta = 0.050$, $P < 0.001$, 95% CI $[0.036, 0.064]$), while left-leaning actors do not differ from baseline ($\beta = 0.001$, $P = 0.945$, 95% CI $[-0.039, 0.041]$). EMI shows a small but statistically robust overall decline over time ($\beta = -0.001$, $P < 0.001$, 95% CI $[-0.001, -0.001]$). This decline is partially offset for right-leaning actors, who show a significantly more positive time trend relative to the centre ($\beta = 0.001$, $P < 0.001$, 95% CI $[0.001, 0.001]$); the interaction with left-leaning actors is not significant ($P = 0.494$). Higher tweet engagement (log like count) is associated with lower EMI ($\beta = -0.040$, $P < 0.001$, 95% CI $[-0.040, -0.039]$), and the lagged EMI term shows strong temporal persistence ($\beta = 0.495$, $P < 0.001$, 95% CI $[0.490, 0.500]$).

For parliamentary speeches (Extended Data Tables A7 and A8, Figure 3 Panel B), the breakpoint regression reveals significant level shifts at each electoral cycle. With breakpoint effects allowed to vary by camp, right-leaning actors exhibit substantially lower baseline EMI ($\beta = -0.347$, $P < 0.001$, 95% CI $[-0.480, -0.213]$); left-leaning actors do not differ from the centre at baseline ($\beta = 0.015$, $P = 0.930$). The 2017 election is associated with a significant downward level shift ($\beta = -0.427$, $P < 0.001$, 95% CI $[-0.485, -0.369]$) without a significant slope change ($P = 0.780$); the camp-specific deviations are not significant. The 2021 election is associated with a further negative level shift ($\beta = -0.218$, $P < 0.001$, 95% CI $[-0.269, -0.167]$) and a negative slope change ($\beta = -0.007$, $P < 0.001$, 95% CI $[-0.009, -0.005]$); the slope decline is further amplified among left-leaning actors ($\beta = -0.010$, $P = 0.006$, 95% CI $[-0.017, -0.003]$). The 2025 election shows a strongly negative level shift ($\beta = -0.490$, $P < 0.001$, 95% CI $[-0.600, -0.381]$) but a significantly positive slope change ($\beta = 0.277$, $P < 0.001$, 95% CI $[0.137, 0.416]$), driven primarily by left-leaning actors ($\beta = 0.757$, $P = 0.001$, 95% CI $[0.309, 1.205]$); these post-2025 slope estimates should be interpreted with caution because of the sparse post-breakpoint data. The lagged EMI term is again positive and significant ($\beta = 0.124$, $P < 0.001$, 95% CI $[0.111, 0.137]$). Equations 2 and 3 impose a piecewise-linear functional form on the

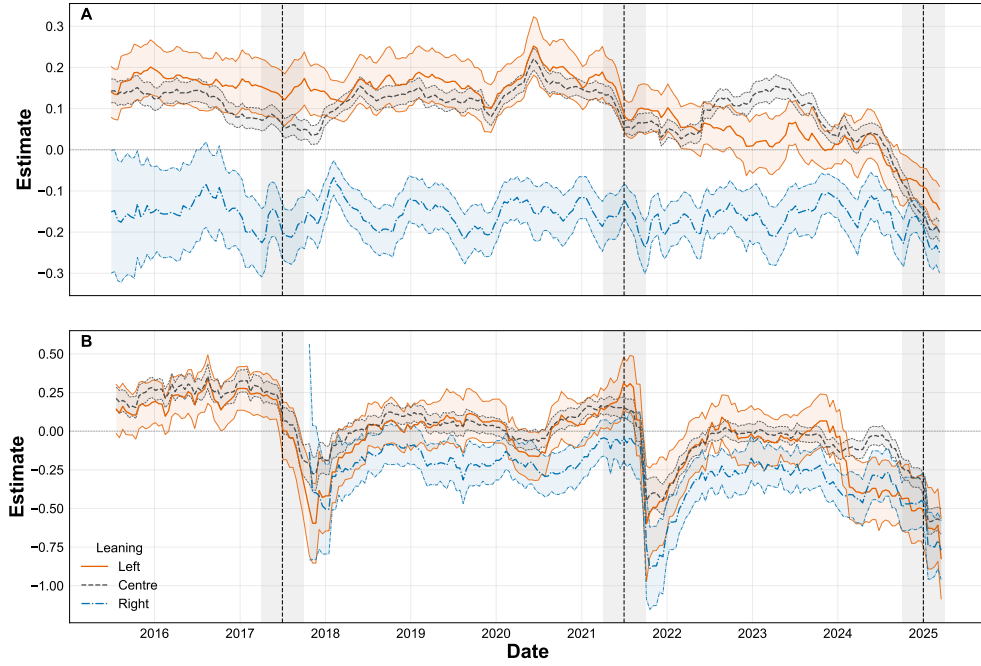


Fig. 4 Rolling-window estimates of evidence-minus-intuition (EMI) scores by ideological leaning on Twitter and in the German Parliament. **A**, Leaning-specific conditional mean EMI on Twitter and **B**, in Parliament, estimated from a linear mixed-effects model fitted within sliding 180-day windows advanced in 14-day steps. Lines show point estimates for Left, Centre and Right; shaded bands are 95% Wald confidence intervals. Grey vertical shading spans one window width around each election breakpoint (1 July 2017, 1 July 2021, 1 January 2025), where window-mean estimates mix pre- and post-breakpoint observations.

EMI trajectories and represent camp differences as constant offsets. To trace those differences without imposing this form, we fitted a simplified linear mixed-effects model within sliding 180-day windows advanced in 14-day steps, entering leaning without a global intercept so that each coefficient estimates that camp's covariate-adjusted mean EMI within the window (See Methods). The resulting trajectories reproduce the before exogenous assumed temporal patterns of the main analysis and make the contrast between the two arenas directly visible in Figure 4. On Twitter (Panel A), the centrist and left-leaning camps decline steadily while right-leaning actors remain close to their initial level, narrowing the gap between the centre and the right over the observation period. In the Bundestag (Panel B), the camps instead step down together at the 2017 breakpoint, with the distance between them largely preserved; narrowing appears only later in the window. In neither arena does the ordering of the camps invert.

3 Discussion

We examine the emergence of extreme right-wing actors in Germany and a concerning epistemic shift in which truth-seeking is increasingly dominated by intuition rather than evidence. Across two complementary arenas of elite political communication in Germany (a strategically curated social-media platform and a highly institutionalised parliamentary chamber), we observe a consistent shift towards intuition-based rhetoric between 2015 and 2025. Right-leaning actors, in particular the AfD, exhibit the lowest EMI values throughout the period, and the broader political discourse co-moves with their increasing visibility. On Twitter this co-movement takes the form of a gradual decline in EMI, consistent with low barriers to participation and faster diffusion of rhetorical conventions [18]. In the Bundestag it manifests as discrete level shifts that align temporally with each electoral cycle, with the most pronounced single shift coinciding with the AfD’s parliamentary entry in 2017. This dual pattern is consistent with two complementary mechanisms of rhetorical convergence: continuous social diffusion online and compositional adaptation within the chamber following the entry of new actors.

These findings extend prior work that documented a long-term decline in evidence-based language in U.S. congressional speeches since the mid-1970s [25] and a parallel cross-country pattern in parliamentary discourse [16, 21]. They also extend the party-competition literature on far-right “contagion.” Prior work documented that mainstream parties respond to radical right success by accommodating its policy positions [26, 27] and by adopting its issue priorities [29, 30]. Our results suggest that contagion may extend to a third dimension that is conceptually orthogonal to both position and topic. we find a chamber-wide and largely party uniformly shift in 2017 shwon by the finding that both mainstream left and mainstream right shift positions once a radical right party enters parliament [27]. Though mainstream parties differ considerably in their formal engagement strategies toward the AfD [50]. The German case is informative on three counts. First, it demonstrates that the shift is not specific to the U.S. two-party system but emerges within a multiparty European democracy. Second, the relatively recent and well-dated entry of a single far-right party into a previously six-party chamber provides a sharper temporal anchor than is available in most multiparty contexts. Third, the cross-platform replication, in two communicative arenas with very different participation rules, suggests the dynamic is not an artifact of a single platform’s affordances. At the same time, the structural breaks at the 2017 electoral cycle are co-temporal with other shocks in our window, notably the COVID-19 pandemic, the Russian invasion of Ukraine, and pronounced changes in social-media platform governance, so we cannot attribute the observed shifts to AfD entry alone. The patterns we report are co-movements consistent with a diffusion or adaptation hypothesis. Stronger causal claims will require designs that exploit exogenous variation in actor visibility (for example, regional discontinuities in AfD success) and explicit network-exposure measures.

Several mechanisms could plausibly underlie the observed convergence towards intuition based language. First, strategic rhetorical mimicry: established actors may adopt elements of the populist rhetorical style in response to perceived voter demand

507 or competitive pressure [51, 52], mirroring the accommodative strategies and posi-
 508 tional contagion documented in the party-competition literature [26–28]. Notably, this
 509 literature implies that even *adversarial* engagement can pull the mainstream onto the
 510 challenger’s terrain, since confronting a niche party raises the salience of its issues [28].
 511 Second, issue-level agenda-setting: when far-right actors successfully shift the topical
 512 agenda toward immigration, identity, and contested values [29, 30], all participants
 513 in the debate must speak about topics that are more amenable to intuition-based
 514 framing. Third, platform amplification: algorithms that prioritise emotionally reso-
 515 nant content can disproportionately surface intuition-based posts, indirectly shaping
 516 the rhetorical environment to which other actors respond [3, 5]. Our data cannot
 517 fully adjudicate between these mechanisms, but two observations are informative. The
 518 contrast between the gradual Twitter decline and the discrete parliamentary shifts
 519 is more easily reconciled with strategic adaptation and issue-level agenda-setting in
 520 the chamber and with diffusion online. Moreover, because the time-trend and break-
 521 point estimates are robust to document-level topic fixed effects, a pure agenda-setting
 522 account (in which the shift reflects only a changed topic mix of the kind documented
 523 during the refugee crisis [29]) cannot fully explain our results: the movement toward
 524 intuition-based rhetoric occurs *within* most topics, indicating a genuinely stylistic
 525 component.

526 Our findings also speak to ongoing debates about the relationship between rhetor-
 527 ical style and democratic deliberation. Evidence-based rhetoric supports a shared
 528 factual baseline that is argued to be necessary for meaningful democratic debate [14],
 529 and prior work has linked declining evidence-based language to higher polarisation
 530 and to the sharing of low-quality information sources [17, 25]. The German shift we
 531 document occurs alongside a documented rise in attacks against members of estab-
 532 lished parties [36, 38] and rising affective polarisation [31, 32]. We do not claim a
 533 causal pathway from one to the other, but the joint pattern motivates closer study of
 534 how rhetorical drift, polarisation, and political violence co-evolve.

535 Our camp-level aggregation necessarily obscures heterogeneity within the left, cen-
 536 tre and right blocs; the centrist camp in particular combines actors with substantively
 537 different rhetorical profiles. Party-level analyses recover the same clustering while pre-
 538 serving these party-specific differences, indicating that aggregation does not distort
 539 the pattern we report, and the position of the most extreme actors is unaffected. Our
 540 analysis is restricted to elite communication and cannot speak to whether comparable
 541 shifts have occurred in mass-public discourse.

542 Future work should test whether network exposure to AfD content predicts within-
 543 actor shifts in EMI for other politicians, an analytic step that would strengthen claims
 544 about diffusion mechanisms beyond the co-movement results we report here. Cross-
 545 country comparisons that exploit different timings of far-right entry into national
 546 parliaments could provide a quasi-experimental test of the convergence hypothesis at
 547 scale.

548 Taken together, our results point to a dimension of elite discourse that changes
 549 alongside far-right visibility and that current accounts of democratic resilience largely
 550 overlook. Mainstream responses to radical right challengers have been studied as con-
 551 tests over positions and issues, and the corresponding remedies, from fact-checking to
 552

corrections and platform moderation, operate at the level of individual claims. What we observe operates at the level of epistemic reasoning. Between 2015 and 2025, the epistemic style of German elite discourse moved towards intuition in both arenas we examined, and it did so most sharply in the Bundestag, the arena with the strongest formal speech norms and the longest standing consensus against the far right. The 2017 shift was chamber-wide, with no significant differences between ideological camps, even though mainstream parties differ considerably in how they formally engage with the AfD [50]. Where the decline of evidence-based language in the U.S. Congress unfolded over roughly half a century [25], comparable movement appears here within a single decade, in discrete steps aligned with changes in the composition of the chamber.

None of this argues for a politics of evidence alone. Intuition carries moral and experiential content that democratic debate cannot do without, and it is over-reliance on it at the expense of evidence that is of concern [21]. But where evidence can no longer adjudicate between competing positions, accountability, informed policy evaluation and cross-partisan deliberation lose their footing [16, 21, 25]. Democracies can withstand deep disagreement about what the facts are. It is less clear that they can withstand disagreement whether facts themselves are a foundational requirement of democratic deliberation.

4 Methods

4.1 Twitter corpus

The Twitter corpus combines two sources. The first is the German subset of the multi-country politician corpus assembled by Lasser and colleagues [16], which was collected by identifying verified Twitter accounts of national politicians and retrieving their tweets via the Twitter API; we use the German subsample from 2008–2021. The second source extends this corpus forward in time by retrieving all tweets posted between 2015 and 2025 from the verified handles of members of the 20th German Bundestag elected in 2022, using the GESIS handle list [45], merged with the handles in the Lasser dataset to maximise coverage of currently and previously serving parliamentarians. Tweets were retrieved between 1 June 2025 and 30 September 2025 via the Twitter (X) Academic API operating under the European Union’s Digital Services Act (DSA, Regulation (EU) 2022/2065) Article 40 access mechanism, which obliges very large online platforms to provide qualified researchers with legally mandated access to publicly-accessible platform data for the study of systemic risks. Where the same tweet was present in both source datasets, the entry was de-duplicated by tweet ID. The final dataset contains 4.5 million tweets from 2,067 politician accounts across 8 parties.

4.2 Parliamentary speech corpora

We use two parliamentary corpora. The *historical corpus* is used solely to pre-train the domain-specific word-embedding model and merges the German Parliamentary Corpus covering 1867–1942 [46, 47] with the Open Discourse Bundestag corpus covering 1949–2021 [48], yielding 329,390 speeches by 3,786 politicians across 22 parties. The

599 *contemporary corpus* is the unit of analysis for the empirical results: it consists of all
600 plenary protocols of the 18th–21st German Bundestag (2015–2025), downloaded from
601 the official Bundestag document server [53]. For each electoral term we retrieved the
602 available XML transcripts and parsed session metadata (date, session number, elec-
603 toral term), speaker information (name, faction, role) and the full text of each speech.
604 Each speech was assigned a unique identifier and linked to its source PDF. The con-
605 temporary corpus comprises 59,170 speeches (after removing procedural speeches) by
606 1,391 politicians across 7 parties.

607

608 4.3 Dictionary translation and validation

609

610 The intuition and evidence dictionaries used in this study were originally developed
611 and validated in English by Aroyehun and colleagues [25]. Briefly, a manually selected
612 seed list by experts (to illustrate, initial keywords for evidence included terms such
613 as “reality”, “assess”, “examine”, “evidence”, “fact”, “truth”, “proof” and so on; for
614 intuition, initial keywords were terms such as “believe”, “opinion”, “consider”, “feel”,
615 “intuition” or “common sense”) was expanded computationally using fastText word
616 embeddings with a cosine-similarity threshold of 0.75 and refined using cross-linguistic
617 colexification networks; overlapping, duplicate and inflected forms were removed, and
618 a human validation study was used to remove or reassign terms that participants did
619 not align with their assigned category.

620 For the present study we translated the validated English dictionaries into German
621 and re-validated them using an online survey on Prolific in which participants scored
622 each candidate term on two five-point Likert scales (representativeness for intuition;
623 representativeness for evidence). Survey data were collected on 21 March 2024 from
624 47 German-speaking participants (35 male, 12 female; mean age 28.9 years, s.d. 7.4).
625 For each term we computed paired *t*-tests of intuition versus evidence ratings, with *p*-
626 values Holm-corrected for multiple comparisons. Of 113 candidate keywords, 86 were
627 retained as significantly aligning with their assigned category; 27 did not reach the
628 corrected significance threshold ($p < 0.05$) and were removed. The final dictionaries are
629 given in Extended Data Table A1; full instructions and demographics of participants
630 are provided in Supplementary Note 2.10.

631

632 4.4 Identification of evidence- and intuition-based components 633 in text

634

635 We use the Distributed Dictionary Representations (DDR) approach, in which a pre-
636 defined dictionary is used to quantify the semantic similarity between a text and a
637 target rhetorical construct via word embeddings, rather than via raw word counts. We
638 trained a domain-specific gensim word2vec model on the historical Bundestag corpus
639 (1867–2025 combined) and fine-tuned the resulting model separately on the Twit-
640 ter and contemporary parliamentary corpora to obtain platform-specific embeddings
641 (training hyperparameters are reported in Supplementary Note 2). The EMI score for
642 each text is computed in three steps.

643 First, for each text *t*, we calculate its distance *D* to the evidence and intuition
644 dictionaries. The distance $D_i(t)$ measures the semantic similarity between the averaged

embeddings of the text and the dictionary:

$$D_i(t) = \text{cos-sim} \left(\frac{1}{|t|} \sum_{w \in t} \mathbf{v}(w), \frac{1}{|i|} \sum_{w \in i} \mathbf{v}(w) \right),$$

where:

- t is the set of words in the input text,
- i is the set of words in the dictionary (*evidence* or *intuition*),
- $\mathbf{v}(w)$ is the embedding vector of word w ,
- $|t|$ and $|i|$ are the number of words in t and i , respectively.

The cosine similarity cos-sim is calculated as:

$$\text{cos-sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|},$$

where:

- $\mathbf{u} \cdot \mathbf{v} = \sum_{j=1}^n u_j v_j$ is the dot product of the two vectors,
- $\|\mathbf{u}\| = \sqrt{\sum_{j=1}^n u_j^2}$ and $\|\mathbf{v}\| = \sqrt{\sum_{j=1}^n v_j^2}$ are their magnitudes.

Next, we z-score normalize the distances to the evidence and intuition dictionaries across all texts. This normalization ensures comparability between scores by centering and scaling the values. Finally, we calculate the EMI score for each text as the difference between the z-scored distances:

$$\text{EMI}(t) = z(D_{\text{evidence}}(t)) - z(D_{\text{intuition}}(t)),$$

where $z(\cdot)$ represents the z-score transformation:

$$z(x) = \frac{x - \mu}{\sigma},$$

with μ and σ being the mean and standard deviation of x , respectively. To analyse trends over time, we calculate the average EMI score for all texts in a given period:

$$\langle \text{EMI} \rangle = \frac{1}{|T|} \sum_{t \in T} \text{EMI}(t),$$

where:

- T is the set of texts in the period,
- $|T|$ is the total number of texts.

This approach allows us to quantify the rhetorical alignment of texts with evidence or intuition-based communication styles over time.

691 4.5 Document-level validation

692 The DDR pipeline was validated through three complementary procedures. First, all
693 EMI scores were recomputed using the publicly available generic German fastText
694 embeddings; the correlation between the gensim-based and fastText-based document-
695 level EMI scores was $r = 0.93$ (95% CI [0.91, 0.94]), indicating that the substantive
696 results are not specific to the fine-tuned embedding model. Second, the contribution
697 of individual dictionary terms to the document-level scores was assessed via 1,000
698 bootstrap resamples of the dictionary composition; document-level scores remained
699 stable across resamples (Supplementary Note 2.8), indicating that no small subset
700 of highly influential terms is driving the observed patterns. Third, we conducted a
701 pre-registered human validation study on Prolific in which 284 German-speaking par-
702 ticipants annotated a stratified sample of 1,300 parliamentary passages (each 10–50
703 tokens long), drawn equally across four EMI bins for each of the three constructs
704 (evidence, intuition, EMI), with no duplicates. We followed the validation procedure
705 of Lasser and colleagues [16]; full instructions are reproduced in Supplementary Note
706 2.11. An attention check (an example item for each construct, repeated within the
707 survey) was used to screen participants; 32 participants failed and were excluded,
708 leaving 252 valid responses. Survey data were collected on 11 February 2025. Single-
709 rater intraclass correlation coefficients (ICC_{single}) were 0.14 (95% CI [0.10, 0.18]) for
710 evidence and 0.16 (95% CI [0.12, 0.21]) for intuition, rising to ICC_{average} values of
711 0.62 (95% CI [0.53, 0.69]) and 0.65 (95% CI [0.57, 0.72]), respectively, when aver-
712 aged across raters. These modest single-rater ICCs are consistent with the inherent
713 ambiguity of short political passages and motivate the use of aggregate-level analyses
714 throughout the paper. To obtain per-dimension ground truth, individual ratings were
715 dichotomised into endorsement (agree or strongly agree) versus non-endorsement, and
716 a chunk was labelled evidence-based (or intuition-based) if a majority of its annotators
717 endorsed that dimension. The EMI ground truth was constructed differently: rather
718 than dichotomising each rating, we averaged the raw ratings per dimension across
719 annotators, took their difference (survey EMI = mean evidence - mean intuition) and
720 labelled a chunk evidence-dominant where this difference exceeded zero. The com-
721 puted scores discriminated the human labels above chance: AUC = 0.68 for intuition
722 and 0.59 for evidence, and 0.72 for EMI against the binarized survey-based EMI (See
723 Supplementary Note 2.11). That the difference score outperforms either component is
724 expected if annotators, like the embeddings, treat the two constructs as partly over-
725 lapping rather than mutually exclusive; the contrast is more reliably identified than
726 either level.

728 4.6 Regression

729 We chose a quarterly data aggregation to reduce noise, capture long-term trends in
730 rhetorical shifts, and align with political and media cycles while ensuring model sta-
731 bility and interpretability. To examine how Evidence Minus Intuition (EMI) scores
732 relate to time and political leaning, and whether German federal elections act as struc-
733 tural breakpoints in those trajectories, we estimated a sequence of three mixed-effects
734 regression models of increasing complexity.
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Baseline model. We first established a baseline by regressing EMI scores on time, political leaning, and their interaction, while controlling for within-actor temporal autocorrelation and stable between-actor differences:

$$\begin{aligned} \text{EMI}_{ij} = & \beta_0 + \beta_1 \cdot \text{time}_i + \beta_2 \cdot \text{leaning}_j \\ & + \beta_3 \cdot (\text{time}_i \cdot \text{leaning}_j) + \lambda \cdot \overline{\text{EMI}}_{j,t-1} + u_j + \epsilon_{ij}. \end{aligned} \quad (1)$$

Here EMI_{ij} is the Evidence Minus Intuition score for document i produced by actor j (e.g., a speech by a parliamentarian or a tweet by a political account); β_0 is the intercept reflecting the baseline EMI score for the reference group at the mean observation time; $\beta_1 \cdot \text{time}_i$ captures the overall linear trend in EMI over time (measured in quarters); $\beta_2 \cdot \text{leaning}_j$ encodes the main effect of an actor’s political leaning (left, centre, or right, with centre as the reference category); and $\beta_3 \cdot (\text{time}_i \cdot \text{leaning}_j)$ allows the temporal trend to differ across ideological groups. The autoregressive term $\lambda \cdot \overline{\text{EMI}}_{j,t-1}$ is the actor’s mean EMI in the quarter preceding document i , which absorbs within-actor serial correlation and ensures that time-trend estimates reflect genuine change rather than carry-over from earlier behaviour. The random intercept $u_j \sim \mathcal{N}(0, \sigma_u^2)$ accounts for stable, unmeasured between-actor differences in baseline EMI, acknowledging, for example, that individual politicians or content creators may differ systematically in their reliance on evidence- versus intuition-based language independently of time or leaning. ϵ_{ij} is the residual error, assumed to be normally distributed.

Breakpoint model. Building on the baseline, we next tested whether EMI trajectories exhibit structural breaks around each German federal election. To do so, we aggregated EMI to quarterly actor-level means $\overline{\text{EMI}}_{jt}$ and introduced piecewise linear regressors for each election year $k \in \{2017, 2021, 2025\}$, with breakpoint $\tilde{b}_k$ set approximately one month before the election date. Time was mean-centred ($\tilde{t}$) to improve numerical stability and interpretability of the intercept:

$$\begin{aligned} \overline{\text{EMI}}_{jt} = & \alpha + \beta_1 \tilde{t} + \beta_2 \cdot \text{leaning}_j + \beta_3 \cdot (\tilde{t} \cdot \text{leaning}_j) \\ & + \sum_k \left[\gamma_k \cdot \max(0, \tilde{t} - \tilde{b}_k) + \delta_k \cdot \mathbf{1}(\tilde{t} \geq \tilde{b}_k) \right] \\ & + \lambda \cdot \overline{\text{EMI}}_{j,t-1} + u_j + \epsilon_{jt}. \end{aligned} \quad (2)$$

The baseline linear trend and leaning effects from Equation 1 are retained. For each breakpoint k , two complementary regressors are added: $\gamma_k \cdot \max(0, \tilde{t} - \tilde{b}_k)$ is a piecewise slope-change term that equals zero before the breakpoint and increases linearly thereafter, capturing a durable shift in the rate of EMI change following an election cycle; $\delta_k \cdot \mathbf{1}(\tilde{t} \geq \tilde{b}_k)$ is a level-shift indicator capturing an abrupt step change in mean EMI at the breakpoint. In this model the breakpoint effects are constrained to be uniform across ideological groups, providing an estimate of the average electoral shock on the EMI trajectory irrespective of leaning.

Full breakpoint model with leaning interactions. Finally, to test whether electoral breakpoints modulate EMI differently for left-, centre-, and right-leaning actors, we extended Equation 2 by fully interacting both breakpoint regressors with

783 political leaning:

$$\begin{aligned}
784 \quad & \overline{\text{EMI}}_{jt} = \alpha + \beta_1 \tilde{t} + \beta_2 \cdot \text{leaning}_j + \beta_3 \cdot (\tilde{t} \cdot \text{leaning}_j) \\
785 \quad & + \sum_k \left[\gamma_k \cdot \max(0, \tilde{t} - \tilde{b}_k) + \delta_k \cdot \mathbf{1}(\tilde{t} \geq \tilde{b}_k) \right. \\
786 \quad & \left. + \gamma_k^\ell \cdot \max(0, \tilde{t} - \tilde{b}_k) \cdot \text{leaning}_j + \delta_k^\ell \cdot \mathbf{1}(\tilde{t} \geq \tilde{b}_k) \cdot \text{leaning}_j \right] \\
787 \quad & + \lambda \cdot \overline{\text{EMI}}_{j,t-1} + u_j + \epsilon_{jt}. \tag{3}
\end{aligned}$$

792 This model is identical to Equation 2 except that γ_k^ℓ and δ_k^ℓ are leaning-specific
793 slope-change and level-shift coefficients, respectively, estimated as deviations from
794 the reference group (centre). A significant γ_k^ℓ indicates that the post-election trend
795 in EMI diverges for a given ideological group relative to the centre; a significant δ_k^ℓ
796 indicates a differential immediate shift in mean EMI at the breakpoint. The lagged
797 outcome $\lambda \cdot \overline{\text{EMI}}_{j,t-1}$ and random intercept $u_j \sim \mathcal{N}(0, \sigma_u^2)$ serve the same purposes
798 as in Equations 1 and 2. All three models were estimated by maximum likelihood
799 (REML = False) using `statsmodels MixedLM`. Model fit was assessed via likelihood-
800 ratio test against an intercept-only null, AIC/BIC, intraclass correlation coefficient
801 $\text{ICC} = \sigma_u^2 / (\sigma_u^2 + \sigma_\epsilon^2)$, and the Durbin–Watson statistic on model residuals to verify
802 that the lagged term adequately addressed residual autocorrelation.

803 **Rolling-window analysis.** Equations 1–3 impose a piecewise-linear functional
804 form on the EMI trajectories. To verify that our conclusions do not depend on this
805 parametrisation, we additionally traced the temporal evolution of EMI with a rolling-
806 window analysis. We slid a 180-day window across each corpus in 14-day steps and,
807 within each window, fitted a simplified linear mixed-effects model regressing document-
808 level EMI on political leaning, the actor’s mean EMI in the preceding quarter and,
809 for Twitter, the log-transformed like count, with a random actor intercept u_j as in
810 Equation 1. Leaning was entered without a global intercept, so that each leaning
811 coefficient directly estimates that group’s covariate-adjusted mean EMI within the
812 window. Windows were estimated only if they contained at least 300 (Parliament)
813 or 500 (Twitter) documents from at least ten actors spanning at least two ideolog-
814 ical groups; all window models were fitted by maximum likelihood, with 95% Wald
815 confidence intervals.

816 4.7 Topic modelling

817 To enable the topic-controlled robustness analysis described below, we
818 estimated a separate topic model for each corpus using BERTopic [54].
819 Documents were embedded with the multilingual sentence-transformer
820 `paraphrase-multilingual-mpnet-base-v2`, which produces 768-dimensional repre-
821 sentations and offers strong German-language coverage. The embedding space was
822 then dimensionally reduced with UMAP and clustered with HDBSCAN under the
823 BERTopic defaults; the resulting clusters were reduced to a target of $K = 30$ topics
824 with a minimum cluster size of 50 documents, a setting chosen to balance topical
825 resolution against the requirement that each retained topic be estimable as a fixed
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effect in the downstream regression. Topics were labelled by their top class-based TF-IDF (c-TF-IDF) terms, and each cluster was then manually inspected and assigned to one of a smaller set of substantive policy domains; this aggregation was used to verify that the retained topics span the main areas of political debate rather than fragmenting a single domain. Full term lists and domain assignments per topic are provided in Supplementary Note 4.

The two corpora differ in size by roughly two orders of magnitude and were fit accordingly. For the contemporary Bundestag corpus ($n \approx 59,000$ speeches), BERTopic was fit on the full corpus. For the Twitter corpus ($n \approx 4.5$ million tweets), BERTopic was fit on a stratified subsample of 500,000 tweets that preserves the joint distribution of ideological leaning and party and the fitted model was then used to transform the full corpus. In both corpora, each document received a hard topic assignment $T_d \in \{-1, 1, \dots, K\}$, with $T_d = -1$ denoting the BERTopic noise cluster of documents that the model could not confidently assign to any coherent topic. Soft topic probabilities were not retained because the downstream fixed-effects design only requires the identity of each document’s assigned topic, not its degree of membership.

4.8 Topic-controlled robustness analysis

To address the concern that observed temporal shifts in EMI could reflect changes in document-level topic composition rather than in epistemic style, we re-estimated each baseline regression with document-level topic fixed effects, using the topic assignments obtained as described above. The largest topic was chosen as the reference category. Each augmented specification adds the same $K - 1$ document-level topic dummies to the corresponding baseline:

$$\overline{\text{EMI}}_{djt}^{(\text{topic-controlled})} = \overline{\text{EMI}}_{djt}^{(\text{baseline})} + \sum_{m=1}^{K-1} \pi_m \cdot \mathbf{1}(T_d = m), \quad (4)$$

where π_m is the mean EMI shift for documents in topic m relative to the reference. The augmentation is applied to the linear baseline (Equation 1) for the Twitter corpus, and to both the constrained and the leaning-interactive breakpoint baselines (Equations 2 and 3) for the Bundestag corpus, with the dispatch matching the baseline that was originally fit to each dataset. Because the FE design absorbs all between-topic variation, the time, leaning and breakpoint coefficients in the augmented models are estimated from *within-topic* variation only, so any shift in the rhetorical signal that is driven by changes in topic mix over time can no longer drive them. Side-by-side comparisons of the baseline and topic-controlled coefficients for each specification are reported in Supplementary Note 4 and confirm that the time-trend, breakpoint, and leaning estimates are robust to topic-mix controls.

Data availability. The validated German evidence and intuition dictionaries, the contemporary Bundestag speech dataset, the tweet IDs of the Twitter corpus (in line with the X Developer Agreement), and all aggregated time-series used in the figures and regression analyses are deposited on the Open Science Framework (OSF) at <https://osf.io/x3zpc/> (DOI to be assigned upon acceptance). The original Lasser tweet IDs are available from Lasser and colleagues [16]. The historical Bundestag corpus is available from Abrami and colleagues [47] and from the Open Discourse project [48]. The contemporary Bundestag XML protocols are publicly available from the Bundestag document server [53]. Access to raw tweet content is restricted by the X Developer Agreement and by the conditions of the DSA Article 40 framework. Researchers vetted to access this dataset under the DSA can have independent access to it, queries about access after approval should be directed to the corresponding author.

Code availability. All code used to construct the corpora, train the embedding models, compute EMI scores, run the validation surveys and estimate the regression models is released under the MIT License on GitHub at <https://github.com/peersal/German-EMI>. Analyses were performed using Python 3.9.1.

Ethics declarations. The two online survey studies were approved by the Ethics Committee of the University of Konstanz (approval no. 03/2023). All participants were recruited via Prolific, were informed of the study’s purpose, and provided informed digital consent prior to participation. Participants were compensated at the Prolific median rate. Tweet content was processed under the conditions of the EU DSA Article 40; only anonymised, aggregated metrics are reported and no personally identifiable information (PII) beyond publicly visible verified-account names of elected politicians is included in any derived dataset.

Competing interests. The authors declare no competing interests.

Author contributions. P.S., S.T.A., S.L. and D.G. designed the study. P.S. collected and curated the corpora, implemented the embedding pipeline and the regression analyses, and ran the document-level validation survey. C.M.A. designed and ran the EMI keyword validation survey, and F.C. conducted the statistical validation analysis of the keyword survey. S.T.A. contributed the DDR pipeline and provided the historical-corpus modelling framework. S.L. provided theoretical input on the evidence/intuition framework. D.G. supervised the project. P.S. wrote the first draft of the manuscript; all authors contributed to revisions and approved the final version.

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Supplementary information. A Supplementary Information document accompanies this submission, covering corpus construction (Note 1), EMI score validation through embedding-model comparisons, dictionary analyses, bootstrapping, two annotator surveys, and a historical validation (Note 2), Regressions (Note 3), and BERTopic diagnostics controlling for agenda-setting (Note 4).

Appendix A Extended Data

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Table A1 Validated German keywords for the intuition and evidence dictionaries.

Intuition Keywords	Evidence Keywords
Ansicht	Akkurat
Bauchgefühl	Analyse
Behauptung	Analysieren
Ehrlichkeit	Auskunft
Fake News	Bildung
Gefühl	Daten
Instinkt	Dossier
Intuition	Ergebnisse
Meinung	Evidenz
Misstrauen	Experte
Perspektive	Fakt
Propaganda	Forschung
Rat	Grund
Ratschlag	Hinweis
Sichtweise	Information
Standpunkt	Intelligenz
Täuschung	Korrektur
Unehrlichkeit	Labor
Verdacht	Logik
Zweifel	Methode
aufrichtig	Nachweis
behaupten	Prozess
beteuern	Prüfung
ehrlich	Punkt
gesunder Menschenverstand	Statistik
glauben	Studie
irreführen	Untersuchung
lügen	Verfahren
offensichtlich	Wahrheit
relativ	Wissen
täuschen	Wissenschaft
unehrlich	echt
unterstützen	erforschen
vermuten	ermitteln
vertrauen	exakt
vertrauenswürdig	gefälscht
verwechselt	genau bestimmen
Überzeugung	inkorrekt
	korrekt
	lernen
	lesen
	logisch
	präzise
	suchen
	untersuchen
	wahr
	wahrheitsgemäß
	wissenschaftlich

Table A2 Illustrative intuition-based passages drawn from the German Twitter and Bundestag speech corpora (2015–2025), shown in the original German with English translation. Passages are sentences from the documents with the highest z-standardised intuition scores in each corpus.

Original (German)	English Translation
<i>Twitter</i>	
“Meinung’ haben Sie schon. . . Aber keine Sorge: ‘Meinung’ aber wenig ‘Wissen’ ist gerade Zeitgeist. . .”	“You do have an ‘opinion’... But don’t worry: ‘opinion’ with little ‘knowledge’ is the zeitgeist right now. . .”
“Alle wissen: wenn man es nur oft genug behauptet, fangen die ersten an, es zu glauben. Dann ist es eine Meinung. Dann wird die Meinung häufiger vertreten, unabhängig von den Fakten [.]”	“Everyone knows: if you just assert it often enough, the first people start to believe it. Then it becomes an opinion. Then the opinion is held more and more widely, regardless of the facts [.]”
“Ich denke, dass er sehr genau weiß was er tut. Ich habe begründeten Verdacht zur Annahme, dass er den Trottel nur spielt [.]”	“I think he knows very well what he is doing. I have well-founded suspicion to assume that he is merely playing the fool [.]”
<i>Bundestag speeches</i>	
“Ich verstehe die Absicht, aber ich vermisse jeglichen Sinn in Ihren Ausführungen. Ich finde es atemberaubend, mit welchen Girlanden und relativ abstrakten Verrenkungen Sie es geschafft haben, uns hier zu erklären, warum wir keinen Ausschluss der Einbürgerung bei Mehr- und Vielehe vornehmen sollten.”	“I understand the intention, but I fail to see any sense in your remarks. I find it breathtaking with what garlands and rather abstract contortions you have managed to explain to us why we should not exclude naturalisation in cases of polygamy.”
“[...] frage ich mich schon: Was sind das für verwirrte Geister, die hier am Rednerpult stehen? Es sind undurchdachte Anträge, die Sie hier vorgelegt haben.”	“[...] I really do ask myself: what kind of confused minds are standing at the lectern here? These are ill-considered motions that you have put forward.”
“Ich kenne Sie als jemanden, der eher juristisch oberflächlich und nicht an Fakten orientiert argumentiert und arbeitet. Das lehne ich schon aus tiefster Grundüberzeugung ab.”	“I know you as someone who argues and works in a legally superficial manner rather than being oriented towards facts. I reject that out of my deepest conviction.”

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Table A3 Illustrative evidence-based passages drawn from the German Twitter and Bundestag speech corpora (2015–2025), shown in the original German with English translation. Passages are sentences from the documents with the highest z-standardised evidence scores in each corpus.

Original (German)	English Translation
<i>Twitter</i>	
“Sie nehmen also eine Studie, welche Ihnen genehm ist, ohne die Annahmen und Methoden zu prüfen und führen das als vermeintlichen Beweis für eine Lüge an? Ich kenne viele Studien dazu, und die meisten haben Mängel.”	“So you take a study that suits you, without checking its assumptions and methods, and present it as supposed proof of a lie? I know many studies on this topic, and most of them have shortcomings.”
“6000 Tote sind nicht erfunden sondern wissenschaftliche Abschätzung auf 100ten von Studien basierend.”	“6,000 deaths are not invented but a scientific estimate based on hundreds of studies.”
“Jede wissenschaftliche Arbeit sollte in Bezug auf Studiendesign und Auswertung von mehr als einer Person begutachtet werden.”	“Every scientific study should be reviewed by more than one person with regard to study design and analysis.”
<i>Bundestag speeches</i>	
“Sie berufen sich hier auf eine Studie von John Cook aus dem Jahr 2013. [...] Schauen wir uns diese Studie einmal an. Was wurde da gemacht? John Cook hat 11 944 wissenschaftliche Arbeiten untersucht.”	“You are relying here on a study by John Cook from 2013. [...] Let us take a look at this study. What was done there? John Cook examined 11,944 scientific papers.”
“Dabei ist der NIPT zunächst nur eine Suche und keine Diagnose. Eine Diagnose erfolgt erst durch eine invasive Fruchtwasseruntersuchung, die das Risiko auf Fehlgeburten erhöht. Eine Vielzahl der Tests liefert falsch-positive Ergebnisse.”	“Yet the NIPT is initially only a screening, not a diagnosis. A diagnosis is only established by an invasive amniocentesis, which increases the risk of miscarriage. A large number of the tests yield false-positive results.”
“Die fehlenden Zitate und nicht belegten Formulierungen verdeutlichen doch ganz klar, dass es sich eben nicht um eine glaubhafte, wissenschaftlich fundierte Analyse handelt [...]”	“The missing citations and unsubstantiated statements make it quite clear that this is not a credible, scientifically sound analysis [...]”

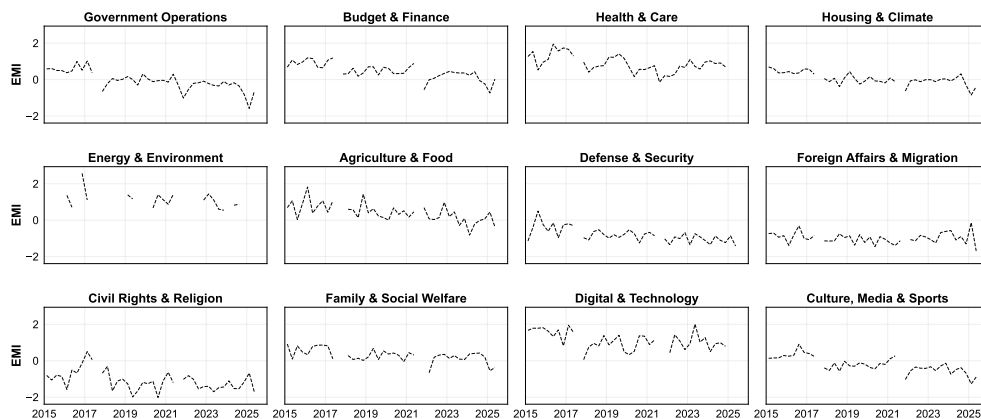


Fig. A1 Evidence-minus-intuition (EMI) scores in the German Parliament by topic, 2015–2025. Each panel shows the quarterly mean EMI of parliamentary speeches assigned to one semantic topic cluster (dashed lines). Topics were obtained with BERTopic ($K = 30$) and aggregated into twelve semantic clusters; documents in the BERTopic outlier cluster were excluded. Quarters with fewer than five speeches in a cluster are left blank. Axes are shared across panels.

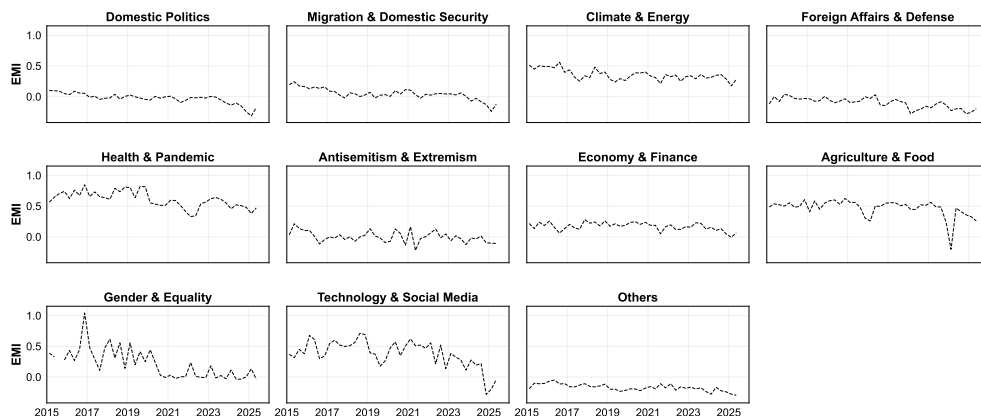


Fig. A2 Evidence-minus-intuition (EMI) scores on Twitter by topic, 2015–2025. Each panel shows the quarterly mean EMI of tweets assigned to one semantic topic cluster (dashed lines). Topics were obtained with BERTopic ($K = 30$) and aggregated into eleven semantic clusters; tweets in the BERTopic outlier cluster were excluded, and topics not assigned to a named cluster are collected under Others. Quarters with fewer than five tweets in a cluster are left blank. Axes are shared across panels.

Table A4 Mean Evidence Minus Intuition (EMI) scores on Twitter by party and ideological leaning, 2015–2025. Values report the unweighted mean across all tweets by all accounts affiliated with each party; standard deviations, standard errors and 95% confidence intervals are derived from the document-level distribution. Rows in italics give the aggregate across all parties of the corresponding leaning, following the Chapel Hill Expert Survey (CHES) classification. The left and right leanings each comprise a single party, so their aggregates equal the respective party values.

	Mean EMI	Std	SE	95% CI	Sample size
The Left Party	0.049	0.734	0.001	(0.047, 0.051)	425246
<i>All left-leaning</i>	<i>0.049</i>	<i>0.734</i>	<i>0.001</i>	<i>(0.047, 0.051)</i>	<i>425246</i>
Alliance 90/The Greens	0.135	0.749	0.001	(0.133, 0.136)	727984
FDP	0.011	0.751	0.001	(0.009, 0.012)	914924
SPD	-0.019	0.763	0.001	(-0.020, -0.017)	608149
CDU/CSU	-0.041	0.752	0.001	(-0.042, -0.039)	590891
<i>All centre</i>	<i>0.026</i>	<i>0.756</i>	<i>0.000</i>	<i>(0.025, 0.026)</i>	<i>2841948</i>
AfD	-0.268	0.688	0.001	(-0.270, -0.266)	508933
<i>All right-leaning</i>	<i>-0.268</i>	<i>0.688</i>	<i>0.001</i>	<i>(-0.270, -0.266)</i>	<i>508933</i>

Table A5 Mean Evidence Minus Intuition (EMI) scores in the German Bundestag by party and ideological leaning, 18th–21st electoral terms (2015–2025). Values report the unweighted mean across all speeches by all members affiliated with each party; standard deviations, standard errors and 95% confidence intervals are derived from the document-level distribution. Rows in italics give the aggregate across all parties of the corresponding leaning, following the Chapel Hill Expert Survey (CHES) classification. The BSW estimate is based on a small sample ($n = 153$) and is reported descriptively only; the right leaning comprises a single party, so its aggregate equals the party values.

	Mean EMI	Std	SE	95% CI	Sample size
The Left Party	0.051	1.268	0.015	(0.021, 0.080)	7038
BSW	-1.098	1.346	0.109	(-1.311, -0.885)	153
<i>All left-leaning</i>	<i>0.026</i>	<i>1.281</i>	<i>0.015</i>	<i>(-0.003, 0.056)</i>	<i>7191</i>
SPD	0.113	1.264	0.011	(0.092, 0.135)	13262
CDU/CSU	0.108	1.258	0.010	(0.089, 0.127)	16690
Alliance 90/The Greens	0.039	1.229	0.013	(0.014, 0.065)	9165
FDP	-0.110	1.187	0.015	(-0.140, -0.081)	6146
<i>All centre</i>	<i>0.066</i>	<i>1.247</i>	<i>0.006</i>	<i>(0.054, 0.077)</i>	<i>45263</i>
AfD	-0.473	1.236	0.015	(-0.502, -0.443)	6716
<i>All right-leaning</i>	<i>-0.473</i>	<i>1.236</i>	<i>0.015</i>	<i>(-0.502, -0.443)</i>	<i>6716</i>

Table A6 Linear mixed-effects regression of Evidence Minus Intuition (EMI) on time, ideological leaning and their interaction for the Twitter corpus (Equation 1). Random intercepts at the actor level. Reference category for leaning is centre.

Model:	MixedLM	Dependent Variable:	emi_w2vparliament
No. Observations:	3707690	Method:	ML
No. Groups:	1063	Scale:	0.5062
Min. group size:	1	Log-Likelihood:	-4000976.3513
Max. group size:	80191	Converged:	Yes
Mean group size:	3487.9		

	Coef.	Std.Err.	z	P> z	[0.025	0.975]
Intercept	0.050	0.007	6.929	0.000	0.036	0.064
leaning[T.right]	-0.111	0.020	-5.620	0.000	-0.149	-0.072
leaning[T.left]	0.001	0.020	0.069	0.945	-0.039	0.041
time.centered	-0.001	0.000	-29.184	0.000	-0.001	-0.001
time.centered:leaning[T.right]	0.001	0.000	20.468	0.000	0.001	0.001
time.centered:leaning[T.left]	-0.000	0.000	-0.684	0.494	-0.000	0.000
log_like_count	-0.040	0.000	-149.524	0.000	-0.040	-0.039
emi_lag1	0.495	0.002	202.166	0.000	0.490	0.500
Group Var	0.038					

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Table A7 Constrained breakpoint linear mixed-effects regression of Evidence Minus Intuition (EMI) on time, ideological leaning and election-year structural breaks (2017, 2021, 2025) for the Bundestag corpus (Equation 2). Breakpoint level shifts (δ_k) and slope changes (γ_k) are constrained to be uniform across ideological camps. Random intercepts at the actor level. Reference category for leaning is centre.

Model:	MixedLM	Dependent Variable:	emi_w2vparliament
No. Observations:	56066	Method:	ML
No. Groups:	1218	Scale:	1.0245
Min. group size:	1	Log-Likelihood:	-81804.4067
Max. group size:	362	Converged:	Yes
Mean group size:	46.0		

	Coef.	Std.Err.	z	P> z	[0.025	0.975]
Intercept	0.623	0.066	9.485	0.000	0.494	0.752
leaning[T.left]	-0.075	0.067	-1.115	0.265	-0.207	0.057
leaning[T.right]	-0.342	0.065	-5.260	0.000	-0.469	-0.214
time_centered	0.003	0.001	2.197	0.028	0.000	0.006
time_centered:leaning[T.left]	0.002	0.001	3.080	0.002	0.001	0.003
time_centered:leaning[T.right]	0.001	0.001	1.217	0.224	-0.001	0.002
slope_elec2017	-0.001	0.002	-0.638	0.524	-0.004	0.002
level_elec2017	-0.421	0.027	-15.787	0.000	-0.473	-0.368
slope_elec2021	-0.008	0.001	-7.287	0.000	-0.010	-0.006
level_elec2021	-0.207	0.023	-9.180	0.000	-0.251	-0.162
slope_elec2025	0.303	0.062	4.883	0.000	0.182	0.425
level_elec2025	-0.502	0.050	-10.061	0.000	-0.600	-0.404
emi_lag1	0.125	0.007	19.036	0.000	0.112	0.138
Group Var	0.390	0.019				

Table A8 Full breakpoint linear mixed-effects regression of Evidence Minus Intuition (EMI) on time, ideological leaning and election-year structural breaks (2017, 2021, 2025) with leaning-specific breakpoint interactions for the Bundestag corpus (Equation 3). Random intercepts at the actor level. Reference category for leaning is centre. Post-2025 estimates are based on a sparse post-breakpoint sample and should be interpreted with caution.

Model:	MixedLM	Dependent Variable:	emi_w2vparliament
No. Observations:	56066	Method:	ML
No. Groups:	1218	Scale:	1.0238
Min. group size:	1	Log-Likelihood:	-81786.6693
Max. group size:	362	Converged:	Yes
Mean group size:	46.0		

	Coef.	Std.Err.	z	P> z	[0.025	0.975]
Intercept	0.607	0.072	8.428	0.000	0.466	0.748
leaning[T.left]	0.015	0.172	0.088	0.930	-0.323	0.353
leaning[T.right]	-0.347	0.068	-5.071	0.000	-0.480	-0.213
time_centered	0.003	0.002	1.652	0.099	-0.000	0.006
time_centered:leaning[T.left]	0.005	0.004	1.341	0.180	-0.002	0.012
time_centered:leaning[T.right]	-0.000	0.002	-0.087	0.931	-0.004	0.004
slope_elec2017	-0.000	0.002	-0.279	0.780	-0.004	0.003
level_elec2017	-0.427	0.030	-14.429	0.000	-0.485	-0.369
slope_elec2017_left	-0.002	0.004	-0.404	0.686	-0.010	0.006
level_elec2017_left	0.018	0.073	0.252	0.801	-0.125	0.162
slope_elec2021	-0.007	0.001	-5.616	0.000	-0.009	-0.005
level_elec2021	-0.218	0.026	-8.387	0.000	-0.269	-0.167
slope_elec2021_left	-0.010	0.004	-2.741	0.006	-0.017	-0.003
level_elec2021_left	0.088	0.075	1.174	0.240	-0.059	0.234
slope_elec2021_right	0.000	0.003	0.050	0.960	-0.006	0.006
level_elec2021_right	0.030	0.064	0.472	0.637	-0.096	0.156
slope_elec2025	0.277	0.071	3.879	0.000	0.137	0.416
level_elec2025	-0.490	0.056	-8.754	0.000	-0.600	-0.381
slope_elec2025_left	0.757	0.228	3.315	0.001	0.309	1.205
level_elec2025_left	-0.493	0.176	-2.802	0.005	-0.838	-0.148
slope_elec2025_right	-0.301	0.178	-1.690	0.091	-0.651	0.048
level_elec2025_right	0.277	0.156	1.774	0.076	-0.029	0.583
emi_lag1	0.124	0.007	18.868	0.000	0.111	0.137
Group Var	0.391	0.019				

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